

WEEK-7 PRACTICAL — Node.js User Service [Pranay Reddy, 2520080061]

Step – 1 Create a User — POST

```
Windows PowerShell
-----
dar--l      21-09-2026    14:48      controllers
dar--l      21-09-2026    14:48      node_modules
dar--l      21-09-2026    14:48      routes
-a---l      21-09-2026    14:44      39264 package-lock.
                                     json
-a-----      21-09-2026    14:44      349 package.json

PS C:\Users\DELL\OneDrive\Documents\S DBE&DBD\Practical\Week 7\n
odememo\user-service> node app.js
User service running on port 5001
MongoDB connected

PS C:\Users\DELL\OneDrive\Documents\S DBE&DBD\Practical\Week 7\n
odememo\user-service> Invoke-RestMethod -Uri "http://localhost:5
001/users" -Method POST -ContentType "application/json" -Body '{
{"username":"Pranay Reddy","email":"Pranay2520080061@klh.edu.in","profile":
{}}}'

message      user
-----
User created successfully @{username=Pranay Reddy; email=2520080061...
```

POST in mongosh:

```
PRANAY REDDY'S Workspace
Overview POST http://localhost:5001/users
http://localhost:5001/users
POST http://localhost:5001/users
Send
Body
raw JSON
1 {
2   "username": "Pranay Reddy",
3   "email": "2520080061@klh.edu.in"
4 }
201 Created 6 ms 386 B
Body Cookies Headers 7 Test Results
{} JSON Preview Visualize
1 {
2   "message": "User created successfully",
3   "user": {
4     "username": "Pranay Reddy",
5     "email": "2520080061@klh.edu.in",
6     "_id": "6ab0f91bf620004aca5e8a07",
7     "__v": 0
8   }
9 }
```

Step 2: Get User — GET

```
PRANAY REDDY'S Workspace
Overview GET http://localhost:5001/users/6ab0f91bf620004aca5e8a07
http://localhost:5001/users/6ab0f91bf620004aca5e8a07
GET http://localhost:5001/users/6ab0f91bf620004aca5e8a07
Send
Body Cookies Headers 7 Test Results
{} JSON Preview Visualize
1 {
2   "_id": "6ab0f91bf620004aca5e8a07",
3   "username": "Pranay Reddy",
4   "email": "2520080061@klh.edu.in",
5   "__v": 0
6 }
```

Output on browser by id:

```
< > ↺ localhost:5001/users/6ab0f91bf620004aca5e8a07
Pretty print
{
  "_id": "6ab0f91bf620004aca5e8a07",
  "username": "Pranay Reddy",
  "email": "2520080061@klh.edu.in",
  "__v": 0
}
```

Step 3: Create Activity — POST

```
PS C:\Users\DELL\OneDrive\Documents\S DBE&DBD\Practical\Week 7\nodememo\user-service> Invoke-RestMethod -Uri "http://localhost:5001/activities" -Method POST -ContentType "application/json" -Body '{"user_id":"6ab0f91bf620004aca5e8a07","action":"SEARCH_PYTHON"}'
```

message activity

Activity created successfully @{user_id=6ab0f91bf620004aca5e...

In mongosh:

Workspace

Overview POST http://localhost:5001/activities

POST http://localhost:5001/activities

Body

```
1 {
2   "user_id": "6ab0f91bf620004aca5e8a07",
3   "action": "SEARCH_JAVA"
4 }
```

201 Created • 9 ms • 439 B

Body

```
1 {
2   "message": "Activity created successfully",
3   "activity": {
4     "user_id": "6ab0f91bf620004aca5e8a07",
5     "action": "SEARCH_JAVA",
6     "_id": "6ab0fb9ef620004aca5e8a08",
7     "timestamp": "2026-09-21T09:40:46.119Z",
8     "__v": 0
9   }
10 }
```

Step 4: Add More Activities

Workspace

Overview POST http://localhost:5001/activities/6ab0f91bf620004aca5e8a07

POST http://localhost:5001/activities/6ab0f91bf620004aca5e8a07

Body

```
1 {
2   "user_id": "6ab0f91bf620004aca5e8a07",
3   "activities": [
4     {
5       "action": "Perform Week 7 Skill"
6     },
7     {
8       "action": "Perform Week 7 Practical"
9     },
10    {
11      "action": "Complete MongoDB Task"
12    },
13    {
14      "action": "Test APIs in Postman"
15    }
16  ]
17 }
```

Body

```
1 [
2   {
3     "_id": "6ab10015f620004aca5e8a0c",
4     "user_id": "6ab0f91bf620004aca5e8a07",
5     "action": "SEARCH_PYTHON",
6     "timestamp": "2026-09-21T09:59:49.278Z",
7     "__v": 0
8   },
9   {
10    "_id": "6ab0fe44f620004aca5e8a09",
11    "user_id": "6ab0f91bf620004aca5e8a07",
12    "action": "Perform Week 7 Practical",
13    "timestamp": "2026-09-21T09:52:04.478Z",
14    "__v": 0
15  }
16 ]
```

```
PS C:\Users\DELL\OneDrive\Documents\S DBE60B0\Practical\Week 7\nodememo> Invoke-RestMethod -Uri "http://localhost:5001/activities/6ab0f91bf620004aca5e8a07" -Method GET

_id      : 6ab10015f620004aca5e8a0c
user_id  : 6ab0f91bf620004aca5e8a07
action   : SEARCH_PYTHON
timestamp : 2026-09-21T09:59:49.270Z
__v      : 0

_id      : 6ab0fe44f620004aca5e8a09
user_id  : 6ab0f91bf620004aca5e8a07
action   : Perform Week 7 Practical
timestamp : 2026-09-21T09:52:04.478Z
__v      : 0

_id      : 6ab0fb9ef620004aca5e8a08
user_id  : 6ab0f91bf620004aca5e8a07
action   : SEARCH_JAVA
timestamp : 2026-09-21T09:40:46.119Z
__v      : 0
```

Output on browser:

localhost:5001/activities/6ab0f91bf620004aca5e8a07

Pretty print ☒

```
[
  {
    "_id": "6ab10015f620004aca5e8a0c",
    "user_id": "6ab0f91bf620004aca5e8a07",
    "action": "SEARCH_PYTHON",
    "timestamp": "2026-09-21T09:59:49.270Z",
    "__v": 0
  },
  {
    "_id": "6ab0fe44f620004aca5e8a09",
    "user_id": "6ab0f91bf620004aca5e8a07",
    "action": "Perform Week 7 Practical",
    "timestamp": "2026-09-21T09:52:04.478Z",
    "__v": 0
  },
  {
    "_id": "6ab0fb9ef620004aca5e8a08",
    "user_id": "6ab0f91bf620004aca5e8a07",
    "action": "SEARCH_JAVA",
    "timestamp": "2026-09-21T09:40:46.119Z",
    "__v": 0
  }
]
```

MongoDB connection:

MongoDB Compass - 127.0.0.1:27017/digital_library

Connections Edit View Help

Compass

127.0.0.1:27017 > digital_library

Collection name	Properties	Storage size	Data size	Documents	Avg. document size
activities	-	32.77 kB	300.00 B	3	100.00 B
users	-	26.86 kB	339.00 B	4	64.00 B

MongoDB Compass - 127.0.0.1:27017/digital_library.activities

Connections Edit View Collection Help

Compass

127.0.0.1:27017 > digital_library > activities

Documents Aggregations Schema Indexes Validation

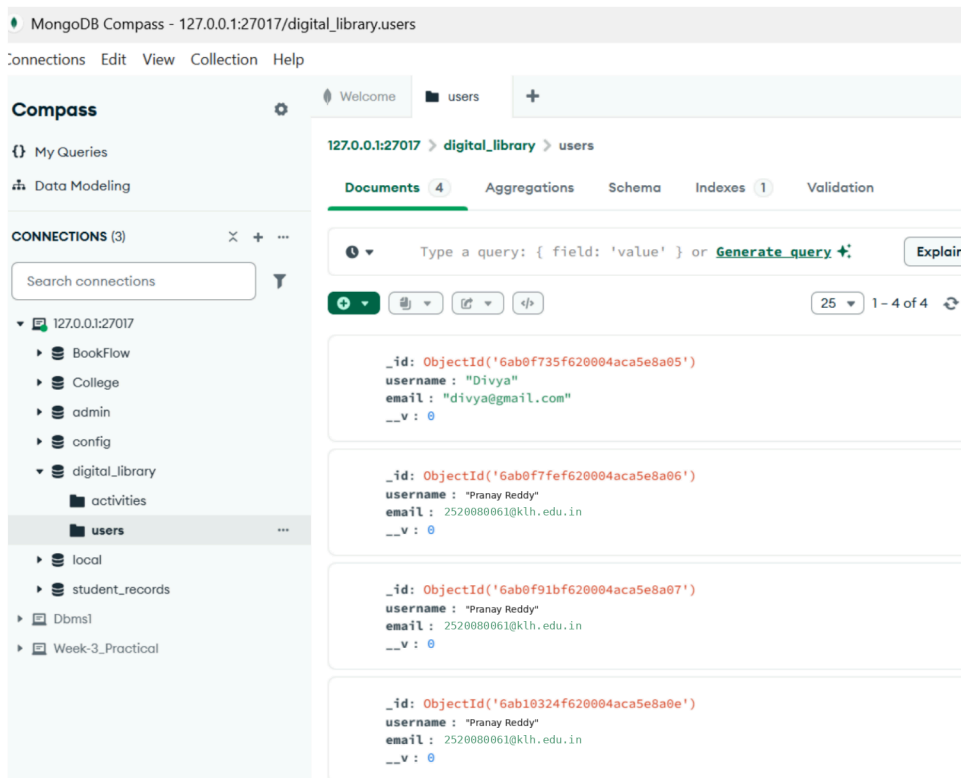
Type a query: { field: 'value' } or Generate query

1-3 of 3

```
{ "_id": "6ab0fb9ef620004aca5e8a08",
  "user_id": "6ab0f91bf620004aca5e8a07",
  "action": "SEARCH_JAVA",
  "timestamp": "2026-09-21T09:40:46.119+00:00",
  "__v": 0 }
```

```
{ "_id": "6ab0fe44f620004aca5e8a09",
  "user_id": "6ab0f91bf620004aca5e8a07",
  "action": "Perform Week 7 Practical",
  "timestamp": "2026-09-21T09:52:04.478+00:00",
  "__v": 0 }
```

```
{ "_id": "6ab10015f620004aca5e8a0c",
  "user_id": "6ab0f91bf620004aca5e8a07",
  "action": "SEARCH_PYTHON",
  "timestamp": "2026-09-21T09:59:49.270+00:00",
  "__v": 0 }
```



Folder Structure:

Week 7

|__nodememo

|__user-service

|

|__controllers

| |__activityController.js

| |__userController.js

|

|__routes

| |__activityRoutes.js

| |__userRoutes.js

|

|__app.js

|__db.js

|__package.json

|__env

db.js

```
const mongoose = require("mongoose");

const MONGO_URI =
  process.env.MONGO_URI ||
  "mongodb://127.0.0.1:27017/digital_library";

mongoose.connect(MONGO_URI)
  .then(() => console.log("MongoDB connected"))
  .catch(err => console.error("MongoDB connection error:", err));

const userSchema = new mongoose.Schema({
  username: {
    type: String,
    required: true
  },
  email: {
    type: String,
    required: true
  },
  profile: {
    type: mongoose.Schema.Types.Mixed,
    default: {}
  }
});

const activitySchema = new mongoose.Schema({
  user_id: {
    type: mongoose.Schema.Types.ObjectId,
    required: true,
    ref: "User"
  },
  action: {
    type: String,
    required: true
  },
  timestamp: {
    type: Date,
    default: Date.now
  }
});

const User = mongoose.model("User", userSchema);
const Activity = mongoose.model("Activity", activitySchema);

module.exports = { User, Activity };
```

controllers/userController.js

```
const { User } = require("../db");

const createUser = async (req, res) => {
  try {
    const { username, email, profile } = req.body;

    const user = await User.create({
      username,
      email,
      profile
    });

    res.status(201).json({
      message: "User created successfully",
      user
    });
  } catch (error) {
    res.status(500).json({
      message: "Error creating user",
      error: error.message
    });
  }
};

const getUser = async (req, res) => {
  try {
    const user = await User.findById(req.params.id);

    if (!user)
      return res.status(404).json({
        message: "User not found"
      });

    res.json(user);
  } catch (error) {
    res.status(500).json({
      message: "Error getting user",
      error: error.message
    });
  }
};

module.exports = { createUser, getUser };
```

controllers/activityController.js

```
const mongoose = require("mongoose");
const { Activity } = require("../db");

const createActivity = async (req, res) => {
  try {
    const { user_id, action } = req.body;

    if (!mongoose.Types.ObjectId.isValid(user_id))
      return res.status(400).json({
        message: "Invalid user_id"
      });

    const activity = await Activity.create({
      user_id,
      action
    });

    res.status(201).json({
      message: "Activity created successfully",
      activity
    });
  } catch (error) {
    res.status(500).json({
      message: "Error creating activity",
      error: error.message
    });
  }
};

const getActivities = async (req, res) => {
  try {
    const { user_id } = req.params;

    if (!mongoose.Types.ObjectId.isValid(user_id))
      return res.status(400).json({
        message: "Invalid user_id"
      });

    const activities = await Activity
      .find({ user_id })
      .sort({ timestamp: -1 });

    res.json(activities);
  } catch (error) {
    res.status(500).json({
      message: "Error getting activities",
    });
  }
};
```

```
        error: error.message
    });
}
};
```

```
module.exports = { createActivity, getActivities };
```

routes/userRoutes.js

```
const express = require("express");
```

```
const router = express.Router();
```

```
const {
    createUser,
    getUser
} = require("../controllers/userController");
```

```
router.post("/users", createUser);
```

```
router.get("/users/:id", getUser);
```

```
module.exports = router;
```

routes/activityRoutes.js

```
const express = require("express");
```

```
const router = express.Router();
```

```
const {
    createActivity,
    getActivities
} = require("../controllers/activityController");
```

```
router.post("/activities", createActivity);
```

```
router.get("/activities/:user_id", getActivities);
```

```
module.exports = router;
```

app.js

```
const express = require("express");
```

```
const app = express();
```

```
require("./db");
```

```
app.use(express.json());
```

```
const activityRoutes = require("./routes/activityRoutes");
const userRoutes = require("./routes/userRoutes");

app.use("/", activityRoutes);
app.use("/", userRoutes);

app.listen(5001, () => {
  console.log("User service running on port 5001");
});
```